# Effects of Acoustic Excitation on a Swirling Diffusion Flame

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*A swirling double concentric jet is commonly used for nonpremixed gas burner application for safety reasons and to improve the combustion performance. Fuel is generally spurted at the central jet while the annular coflowing air is swirled. They are normally separated by a blockage disk where the bluff-body effects further enhance the recirculation of hot gas at the reaction zone. This paper aims to experimentally investigate the behavior of flame and flow in a double concentric jet combustor when the fuel supply is acoustically driven. Laser-light sheet assisted Mie scattering method has been used to visualize the flow, while the flame lengths were measured by a conventional photography technique. The fluctuating velocity at the jet exit was measured by a two-component laser Doppler velocimeter. Flammability and stability at first fuel tube resonant frequency are reported and discussed. The evolution of flame profile with excitation level is presented and discussed, together with the reduction in flame length. The flame in the unforced reacting axisymmetric wake is classified into three characteristic modes, which are weak swirling flame, lifted flame, and transitional reattached flame. These terms reflect their primary features of flame appearances, and when the acoustic excitation is applied, the flame behaviors change with the excitation frequency and amplitude. Four additional characteristic modes are identified; e.g., at low excitation amplitudes, wrinkling flame with a blue annular film is observed because the excitation induces vortices in the central fuel jet and hence gives rise to the wrinkling of flame. The central jet vortices become larger with the increase in excitation amplitude and thus lead to a wider and shorter flame. If the excitation amplitude is increased above a certain value, the central jet vortices change the rotation direction and pacing with the annular jet vortices. These changes in the flow field induce large turbulent intensity and mixing and therefore make the flame looks blue and short. Further increase in the excitation amplitude would lift the flame because the flow field would be dramatically modified.* [DOI: 10.1115/1.4001768]

*Keywords: reacting swirling wake, acoustic excitation, nonpremixed flame*

## 1 Introduction

Swirl flow has been employed to enhance mixing and combustion for industrial gas burners suited for boilers, gas furnaces, gas turbines, etc. The tangential velocity component imparted by the swirling flow causes a low relative pressure at the central axis and thus triggers recirculation at the central axis, which expedites mixing if fuel is spurted as a central jet. However, the swirl number has to be beyond 0.6, or an annular jet Reynolds number has to be more than 18,000 to make recirculation possible [1]. Many of the modern day industrial gas burners, particularly for small to medium scale purposes, operate at much lower swirl and Reynolds numbers.

Downstream recirculation naturally occurs behind a firmly mounted blockage disk along the flow stream at the right Reynolds number. Behind such a disk is a subatmospheric pressure that will entice a recirculation of fluids called the bluff-body effect. But bluff body offers considerable pressure loss to the fluid flow, particularly when the blockage size becomes large. A recirculation bubble was observed by Escudier and Keller [2] immediately behind the cylindrical center body with a recirculation zone at the downstream area. The wake is enclosed in a cylindrical tube where the center bodies' blockage ratios are 0.25 and 0.39. Huang and Tsai [3] used 14 blockage disks to investigate the recirculation of a swirl flow with swirl number ranges from 0.02 to 0.6. The blockage ratio ranges from 0.026 to 0.563, where the bluff-body effects were found significant if the blockage ratio was more than 0.11. Several flow structures were found, such as *single bubble*, *dual rings*, and *dual rings* with *vortex breakdown*. Huang and Tsai [4] made a follow-up investigation particularly on the mixing capability coupled with laser Doppler velocimeter (LDV) measurements. They concluded that larger central blockage requires least swirl strength to attain better mixing. However, mixing is not satisfactorily high, particularly with a dual ring flow structure because the central jet is entangled by the jetlike swirling flow at the central axis.

An alternative way to improve the entrainment of surrounding air into the combusting jet is by pulsing the fuel supply. Jet pulsing is commonly done either by acoustic excitation [5–15] or by a valve switching mechanism [16–18]. Acoustic excitation is carried out by mounting a loud speaker along the fuel supply line, while valve switching usually employ a solenoid valve or a rotating perforated disk to periodically cut the fuel supply. Either way, investigators reported an amplified jet exit velocity fluctuation especially when the pulsation is tuned close to the fuel tube resonant frequency [5,7–11,15–17]. Crow and Champagne [5] defined a preferred mode of excitation at a Strouhal number of 0.3 by which the jet response is a maximum velocity fluctuation amplitude. At an excited central jet Strouhal number of 0.4, Strawa and Cantwell [6] discovered a strong coupling with the coflowing annular jet by which the luminous zone pinches off. Their schlieren images trace the deceleration of the central reacting jet at such pinch-off point. The large fluctuating amplitude of the pulsing jet tuned at the corresponding fuel tube resonant frequency exhibits a dramatic reduction in flame length [7–9,16]. Pulsing a confined jet at a confinement tube resonant frequency, Parikh and Moffat [17] re-

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ported that the entrainment at least tripled compared with the unpulsed jet. Acoustic excitation was also proven as a powerful tool to control flame lifting [11,12,14] and flame blow out [15]. The enhanced premixing during acoustic excitation on a lifted flame is a big help to reduce $NO_x$ emission [14]. As the fuel jet pressure varies during the cycle of flow forcing, the fuel outward vortices collapse into the entrained flow named by Kim et al. [7] as the phenomena of *collapsible mixing*. Some of their co-workers [9] made a follow-up investigation on the behavior of flame at feeble forcing or minimal excitation voltage. It signifies that a burning jet can be affected by propagating sound waves even at low amplitudes. They also found out that in the course of increasing acoustic forcing amplitude, the fuel vortices change their direction and cause the flame to elongate. The work of Demare and Baillot [10] concluded that the acoustic excitation transformed the long and yellow soot radiating flame into a short blue and stable flame. However, improper acoustic forcing also offers detrimental combustion effects on the likes of flame extinction, flash back, poor combustion, and system efficiency reduction. Baillot and Demare [11] made a classification on the behavior of flames at various forcing frequencies and fuel jet velocities where some illustrate the adverse effect on combustion.

Nonpremixed combustors are usually involved in industrial application for safety reasons in exchange of the low combustion efficiency due to poor mixing. The present combustor is one of them, as described by Huang and Tsai [4]. It is a double concentric combustor where the fuel jet is spurted at the central axis with an annular swirling air separated by a blockage disk. Inefficient burning of fuel from this combustor may give rise to carbon monoxide poisoning; thus, improved mixing at the wake is a paramount importance. Huang and Yen [19] diverted the central jet fuel to the annular rotating vortex by installing a control disk at the fuel jet exit. Indeed, turbulence intensity was amplified, mixing was improved, and better combustion was attained manifested by a blue flame. The present work aims to enhance mixing at the swirling wake by an acoustic excitation of the central fuel jet. The entrainment of annular swirling air into the pulsing fuel is something that was rarely found in the literature. Our specific objectives are (i) to find the best combination of parameters by which better combustion is achieved and (ii) to describe the flame behavior and mixing phenomena of flow for some of those chosen parameters.

## 2 Experimental Arrangement

The arrangement of the experimental setup is shown in Fig. 1. A 3.4 kW ring blower with a maximum static pressure of 98.1 bars is used to provide the annular swirling air flow. To minimize the flow perturbation caused by the rotating blower blades, acoustic filters are used. Air flows are quantified using rotameters before they are fed into the test rig. The rotameters are calibrated using a microflow calibration system, and the accuracy is within 1% at full scale. The test rig is a $600\times400$ mm$^2$ circular cylinder with an internal cavity profiled into a nozzle where the contraction ratio is 9. At the test rig, quantified air flow passes through a settling chamber, honeycomb, and mesh screen and is finally redirected by 12 axis-symmetric guide vanes to generate swirl flow. Guide vanes are profiled with NACA 0012 having chord and span lengths of 60 mm and 50 mm, respectively. Arranged on a pitch circle of radius 175 mm, the vane angle can be varied from 0 deg to 70 deg with a minimum resolution of 0.5 deg. Shown in the subset of Fig. 1 is the test section located at the exit of the test rig where a 1 mm thick blockage disk is positioned. The exit diameter $D_o$ is 40 mm, while the blockage disk diameter $D$ is 30 mm, making a blockage ratio $B=(D/D_o)^2$ equal to 0.563.

The central jet is commercial grade propane, which is composed of 95.0% $C_3H_8$, 3.5% $C_2H_6$, and 1.5% $C_4H_{10}$. This fuel is stored in an adjacent pressure tank, conveyed by a flexible high pressure hose where the line pressure can be efficiently controlled by a pressure regulator. The fuel gas goes through needle valves

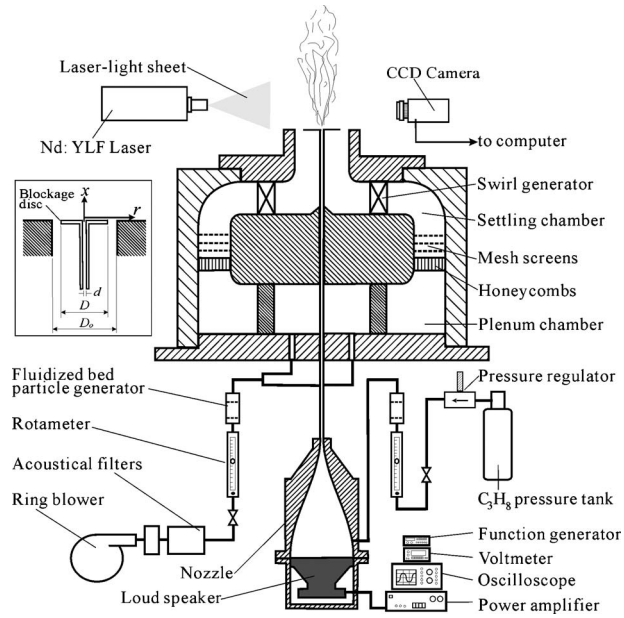


**Fig. 1 Experimental setup**

prior to flow measurements by rotameters. An acoustic excitation arrangement by placing the loud speaker at the base of the fuel tube, as shown in Fig. 1, is adopted for the interaction of sound waves with the fuel jet. This arrangement can also be found in Ginevsky et al. [20]. The 300 mm length nozzle is profiled into a fifth order polynomial where the contraction ratio is 900. A seamless stainless steel tube with a length of 600 mm and an inner diameter of 5.0 mm is connected to the nozzle exit and the blockage disk. Fuel jet fluid is vented to the wake of the blockage disk via a 5.0 mm diameter hole ($d$=5.0 mm) located at the center of the blockage disk. Acoustic excitation is carried out by a 15 cm face diameter loud speaker. The electrical resistance of the voice coil is 6.6 $\Omega$ measured at the loud speaker terminals. The moving mass of the loud speaker, which includes the voice coil, dust cap, surround, spider, and diaphragm, is 22.07 g. A square wave signal is generated by a function generator, and the output signal is verified by an oscilloscope. The frequency at the function generator is varied from 10 Hz to 1200 Hz with a minimum resolution of 1 Hz. The output voltage from the function generator is amplified by a power amplifier. Such amplified signal oscillates the voice coil of the loud speaker, and the maximum attainable excitation voltage is 25 V. A multimeter is used to measure the root mean square voltage across the terminal of the loud speaker.

Image acquisitions are carried out by three different cameras intended for three specific purposes. A Sony progressive scan camera (DCR-TRV 900NTSC) is used to capture colored moving pictures for illustrative purposes. IDT X-Stream Vision high speed camera model XS-4 is used to capture a black and white moving picture to investigate the evolution of vortices. Nikon still camera (Nikon COOLPIX 990) is used for conventional photography technique. The detailed operation of each of these image acquisition tools will be discussed in the following sections.

Jet velocities were measured with a two-component LDV, while the flow is seeded with 1 $\mu$m diameter MgO particles. According to the approximation of Hjelmfelt, Jr. and Mockros [21], the response frequency of these particles in a propane medium is 9000 Hz. The blue and green laser beams, which were supplied by a 6 W argon-ion laser (Spectra-Physics' Stabilite-2017), were separated and focused through the Dantec Fiber-Flow optical system. Typical properties of the optical configurations were listed in Table 1. The "X" configuration for the two-component light beams was used. Such arrangement improved the measurement



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**Table 1 Properties of LDV optical configurations**

| | Green | Blue |
|---|---|---|
| Wavelength, $\lambda$ (nm) | 514.5 | 488 |
| Focal length of receiving and transmitting lens (mm) | 500 | 500 |
| Beam intersection angle (deg) | 8.464 | 8.464 |
| Diameter at beam crossing (mm) | 0.234 | 0.222 |
| Fringe separation ($\mu$m) | 3.486 | 3.306 |
| Number of fringe at measuring volume | 67 | 67 |
| Optical shift frequency (MHz) | 40 | 40 |
| Dimension of measuring volume, $\delta_x \times \delta_r \times \delta_z$ (mm) | $0.235 \times 0.234 \times 3.170$ | $0.223 \times 0.222 \times 3.007$ |

accuracy as both components yield the same values of the jet exit velocities at the center. A Bragg cell was embedded in the system to distinguish the directional ambiguity [22]. A Dantec flow velocity analyzer (FVA) enhanced two-component correlation processor was used to convert the Doppler signals into frequency data. The frequency data were fed into a PC-controlled data acquisition system to calculate the velocities and other statistical properties. The probe is securely mounted on a traversing mechanism where the minimum resolution is 10 $\mu$m.

## 3 Results and Discussion

**3.1 Flame Modes Without Acoustic Excitation.** The swirl strength of the annular jet is defined by swirl number $S$, a dimensionless ratio between the axial fluxes of angular momentum to the axial fluxes of axial momentum [1]. From the time-averaged axial $u$ and azimuthal $w$ velocity components of the annular flow, the swirl number can be obtained using

$$S = \frac{\int_{D/2}^{D_o/2} uwr^2 dr}{\frac{D_m}{2}\int_{D/2}^{D_o/2} u^2 r dr} \tag{1}$$

where $D_m$ is the mean diameter ($=(D+D_o)/2=35$ mm) of the annular jet exit. Measurements and analysis done by Huang and Tsai [3] on the present geometry of the test apparatus showed that the swirl number increases almost linearly with the annular Reynolds number for $\mathrm{Re}_a < 400$. However, when $\mathrm{Re}_a$ is greater than 400, the increasing rate deviates slightly from linearity, which describes a basic and fundamental feature of a guide-vane swirl generator [1]. Annular jet Reynolds number $\mathrm{Re}_a$ is the exit Reynolds number of annular flow defined by

$$\mathrm{Re}_a = \frac{u_a D_h}{\nu_a} \tag{2}$$

where $u_a$ is the volumetric mean axial velocity of the annular swirling air jet at the exit, $D_h$ is the hydraulic diameter of the annular jet at the exit, and $\nu_a$ is the air's kinematic viscosity. Similarly, the central jet Reynolds number $\mathrm{Re}_c$ is calculated from

$$\mathrm{Re}_c = \frac{u_c d}{\nu_f} \tag{3}$$

where $u_c$ is the volumetric mean velocity of the fuel jet, $d$ is the fuel jet exit diameter defined previously, and $\nu_f$ is the fuel's kinematic viscosity.

In the case that no annular swirling air flow is supplied (which is not shown here), i.e., $\mathrm{Re}_a=0$, it looks like a typical laminar jet diffusion flame with a blue flame appeared near the exit region. As the jet diffusion flame is imparted by swirling annular air flow, the flame shape is modified. Typical flame appearances are shown in Fig. 2(*a*) for $\mathrm{Re}_c=2386$. It exhibits three distinguishable appearances, which can be found in different regimes characterized by annular air swirl strength and central fuel jet Reynolds number. As shown in Fig. 2(*b*), at minimal swirl strength ($S=0.316$, $\mathrm{Re}_a=364$), the bottom area of the flame slightly swells compared with that of jet diffusion flame. The region where the flame anchors looks similar to the flame in the wake behind a bluff body [1–4,19] that grows with the increase in swirl strength. It is always accompanied by flame shortening [1,19], which is also illustrated in Fig. 2(*b*). A tarnish and blue flame apparently attaches the base, which extends to $0.4D$. This is the effect of flame stretching at the base that produces excessive strain and pushes the flammable layer beyond the lean combustible limit, which generates local flame extinction [11,12,14]. Further increase in swirl strength to 0.541 ($\mathrm{Re}_a=1000$) lifted the flame, as shown in Fig. 2(*a*). Even though the flame is already lifted, its length persists to decrease in all cases of central jet Reynolds number, as summarized in Fig. 2(*b*). Flame lifting is a common jet diffusion flame phenomenon achievable by increasing the fuel velocity [11,12,14]. But that is not the reason for this kind of flame lifting, which is influenced by the annular swirling jet. It is the fuel and air vortices that move to the same direction, causing poor fuel dispersion [19]. The resulting mixture is not enough to support combustion at the central jet exit; that is why the flame is lifted. This mechanism of flame lifting offers a rare opportunity of premixing in contrast to other works [11,12,14]. At the maximum swirl strength ($S=0.596$, $\mathrm{Re}_a=1700$), the base of the lifted flame moves upstream and reattaches to the blockage disk, as revealed

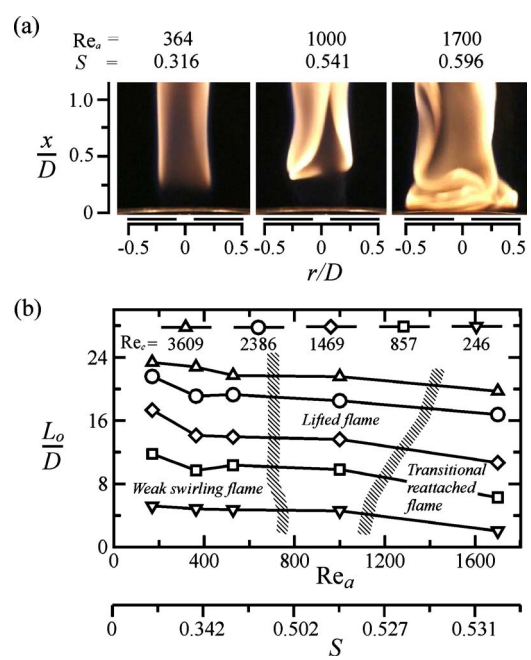


**Fig. 2 Characteristic modes of the unexcited flame at swirling wake. (*a*) Shutter speed, 1/1000 s, $\mathrm{Re}_c=2386$. (*b*) Dimensionless flame length in the domain of annular swirl strength.**



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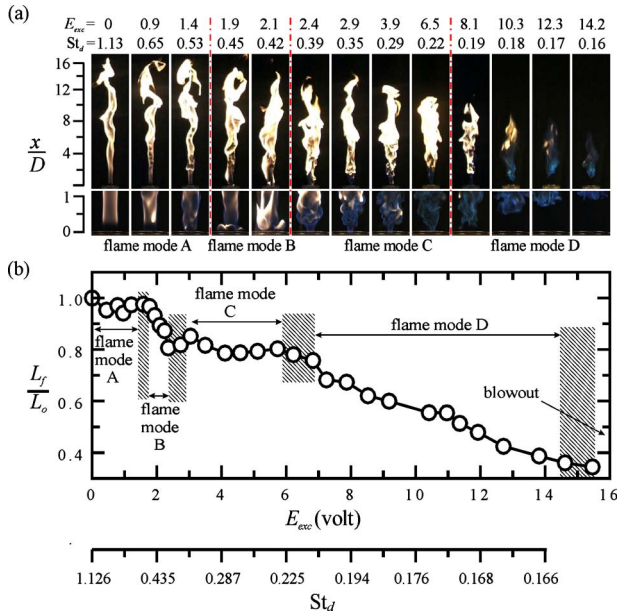


**Fig. 3 Flame behaviors during acoustic excitations: $Re_c$ =2386, $S$=0.316 ($Re_a$=364), and $f$=180 Hz. (*a*) Far field images; shutter speed varies from 1/350–1/725 s. Near field images varies from 1/500–1500 s. (*b*) Dimensionless reduction of flame length in the domain of jet exit Strouhal number corresponding to the excitation level.**

in Fig. 2(*a*). The short but soot radiating yellow flame, which fully occupied the top of the blockage disk, signifies a strong recirculation caused by high annular swirl flow. The flame base remains yellow, but this is the shortest of all flames classified so far, which can be seen at the rightmost side of Fig. 2(*b*). Huang and Yen [19] made a similar classification except for *transitional reattached flame*, which they named as *turbulent reattached flame*. The two other classifications remain the same, such as *weak swirling flame* and *lifted flame*. Flame lengths were measured by a conventional photography technique. A Nikon still camera (Nikon COOLPIX 990) was used to take ten images of the flame with a 1 s exposure time. This means that for all flame length measurements of the present work, each data was gathered from a total of 10 s exposure time.

**3.2 Flame Modes With Acoustic Excitation.** Acoustic excitation triggers the "natural flame" discussed in Sec. 3.1 to exhibit several modified flame behaviors at various excitation levels. As shown in Fig. 3, they are categorized into four groups: flame modes A, B, C, and D. Apparently, the excitation voltage dictates the modification of the central jet Strouhal number at the fuel tube exit. The Strouhal number is calculated as

$$\mathrm{St}_d = \frac{fd}{u_{\mathrm{rms}}} \tag{4}$$

where $f$ is the excitation frequency, $d$ shown above is the fuel tube exit diameter, and $u_{\mathrm{rms}}$ is the root mean square value about the time-averaged velocity of the jet during excitation. Excitation frequency is tuned at 180 Hz, while the central jet Reynolds number is 2386 and the annular swirl strength is 0.316 ($\mathrm{Re}_a$=364). The reason behind choosing such excitation frequency and flow conditions will be discussed in detail in Sec. 3.3, together with the root mean square ($u_{\mathrm{rms}}$) velocity at the fuel tube exit.

As shown in the unexcited flame of Fig. 3(*a*), apparent flame twisting is visible from 3 to 8$D$ due to annular swirl. Previous work carried out by Huang and Tsai [23] on the present flow condition reported that a dual ring with vortex breakdown [1–3] exists at the upstream of such a twisting zone. A rapid drop in the jet exit Strouhal number from 1.126 corresponding to an unexcited case down to about 0.5 is obtained as excitation commenced. As shown in Fig. 3(*a*), at a jet exit Strouhal number of 0.65 ($E_{\mathrm{exc}}$=0.9 V), the acoustic excitation makes the flame base wrinkle. An annular blue flame surrounds the core region of yellowish flame. Wrinkling with a blue annular film near the flame base characterizes flame mode A, which shows a somewhat increased combustion capability compared with the corresponding unexcited case of Fig. 2(*a*). The transition period from flame mode A to B is manifested by a strong coupling [6] between the excited jet and the annular flow. Periodic flame structures pinch off from the wrinkled flame surface through the outer annular film and move upstream as excitation slightly increases. Severe violent wrinkling motions are very obvious in the downstream area of flame mode B accompanied by a very obvious nonuniform bulk expansion around the root area. Oh and Shin [8] spotted seven symmetric azimuthal structures surrounding the central jet during acoustic forcing. A similar observation was reported by Demare and Baillot [10] on such azimuthal structures, which they called *filament*. The highest flame length reduction with a slight change in Strouhal number in flame mode B, as shown in Fig. 3(*b*), is in good agreement with the work of Crow and Champagne [5]. A strong response of the jet exit velocity amplitude to a minimal change in excitation level exists in this flame mode. Increasing the central jet pulsing amplitude pushes the bulky structure at the root area to the downstream section, leaving a completely blue anchored flame at the burner tip. Such behavior is defined as flame mode B, where the flame length continues to drop with the jet exit Strouhal number. A typical flame appearance of flame in this mode is shown in Fig. 3(*a*) for the jet exit Strouhal number less than 0.39 ($E_{\mathrm{exc}}>2.4$ V). The flame exhibits a blue diverging base and looks like an inverted conical flame, which burns a considerable amount of mixtures at the base, resulting to a much shorter flame. Seemingly blue pebbles pinch off at the trailing edge of soot radiating yellow flames. Severe strain at the burner base causes the flame to lift [11,12,14] as excitation is increased to a much higher level (Fig. 3). The lifted flame's leading edge is blue and exhibits strong turbulent motion, which is characterized as flame mode D. This kind of flame behavior always happened when the Strouhal number and swirl strength are pushed down to their minimum values. Persistent and increasing excitation amplifies the lifted flame oscillation while receding to a far downstream area until it never comes back and then blows out [15].

**3.3 Characteristic Regimes.** Similar appearances of flame during acoustic excitation are mapped in Fig. 4(*a*) in the domain of excitation level and frequencies. The narrow band of boundaries represents uncertainty and hysteresis during the transition period from one flame mode to another mode. There exists a threshold of excitation amplitude for each excitation frequency. Exciting the flame with an amplitude lower than the threshold would cause no effect on the flame appearance. Throughout the range of excitation frequencies, flame modes A, B, C, and D appear in an order of increasing excitation level. It means that flame mode C always has the higher range of excitation level than flame mode B for the same excitation frequency. If the excitation frequency is equal or close to 180 Hz, or its harmonics, a minimal amount of excitation level transforms the flame modes due to strong fuel jet oscillations caused by fuel tube resonance [24]. Dramatic reduction in flame length was reported by several investigators [5,7–11,15–17] of the axisymmetric jet flame at the fuel tube resonant frequency.

Logically, the first fuel tube resonant frequency is considered in the present work to investigate the best performance of the system to improve the combustion. The central jet exit velocity was measured by a two-component LDV only at the first fuel tube resonant frequency. Better repeatability was obtained with at least 30,000 samples collected with a maximum data rate of 4.2 kHz. The volumetric deviation of the calculated flow rate at the jet exit varies from 0.08% to 0.05% compared with the rotameter reading



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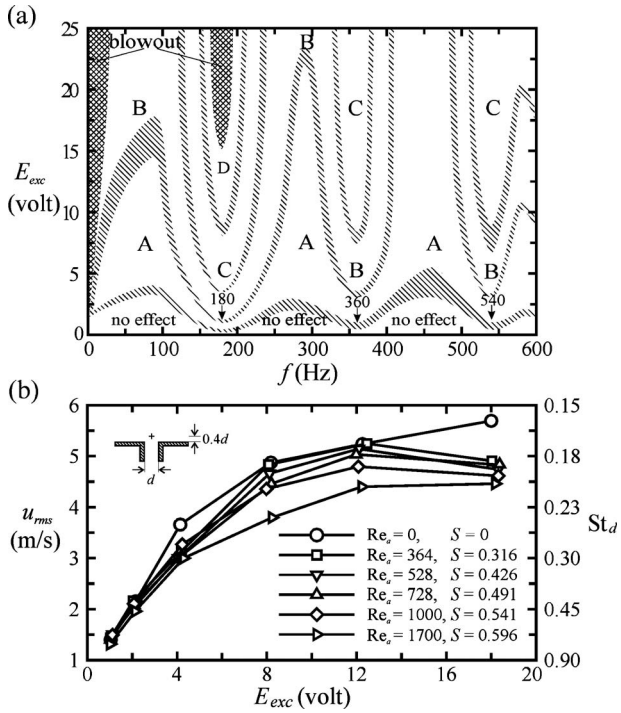


**Fig. 4 Re$_c$=2386. (*a*) Regimes of each flame modes in the domain of excitation level and frequency. *S*=0.316, Re$_a$=364. (*b*) Root mean square of velocity fluctuations at the central jet exit excited at the first fuel tube resonant frequency (*f*=180 Hz).**

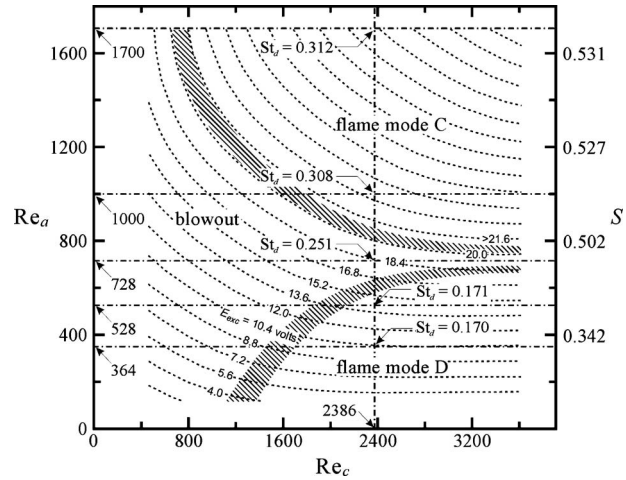


**Fig. 5 Flammability and stability chart at first fuel tube resonant frequency (*f*=180 Hz)**

of the fuel supplied. As shown in Fig. 4(*b*), the root mean square velocity of the fluctuating jet during acoustic excitation reduced with the increased swirl number. These observations illustrate that the recirculation at the wake will have a damping effect on the fluctuating central jet. Off-resonant frequencies should have similar velocity fluctuation behaviors for the same flame mode. The exponential profile of the plot is a limiting factor that describes the behavior of the acoustic generator. Using Eq. (4), the Strouhal number corresponding to the acoustic excitation voltage is calculated based on the 180 Hz fuel tube resonant frequency and other parameters introduced previously. A rapid increase in velocity fluctuation amplitude is attained near the Strouhal number of 0.3, which Crow and Champagne [5] described as their preferred mode. The second and third fuel tube resonant frequencies both yield strong responses, as shown in Fig. 4(*b*), but definitely have different relationships with the jet exit Strouhal numbers and excitation voltages. The range of excitation voltages of such off-resonant flame modes vary, although the range of Strouhal numbers remains the same for a specific mode of flame, which is directly influenced by the jet exit velocity amplitude [5,13]. This can be validated by taking a closer look at Figs. 5 and 6 as the range of Strouhal numbers are very consistent on various flame modes at different swirl strengths. A jet exit Strouhal number less than 0.5 is hardly achieved by those off-resonant frequencies, particularly at the acoustic antinode [24]. For example, at $f=270$ Hz only the transition between flame modes A and B exists at the maximum excitation level. Such transition from flame mode A to B happened at a jet exit Strouhal number of about 0.5, as shown in Fig. 3(*b*).

Figure 5 shows the characteristic modes and stability limit of swirling flame excited at the first fuel tube resonant frequency in the domain of central jet Reynolds number and annular swirl strength. Dashed lines indicate the root mean square excitation voltage measured at the loud speaker terminals. Jet exit Strouhal numbers are only found at the central jet exit Reynolds number of 2386. For example, if the central jet Reynolds number is 2386 and the swirl strength is 0.491 (Re$_a$=728), the flame blows out if the Strouhal number is 0.251 ($E_{\text{exc}}$=18.4 V). For flame mode D on the lower right hand area of Fig. 5, dashed line numbers indicate the approximate excitation level where a stable lifted flame starts to exist. For instance, if the swirl strength is 0.316 (Re$_a$=364) and the central jet Reynolds number is 2386, flame lifting exists when the Strouhal number is 0.170 ($E_{\text{exc}}$=10.4 V). Usually, when the excitation level of flame mode D drops below the Strouhal number of 0.16 ($E_{\text{exc}}>20$ V), the flame blows out at any combination of central and annular jet Reynolds number studied in this work. The upper right hand area of Fig. 5 is the vicinity of strong and stable flame mode C where dashed line voltages guarantee neither unstable flame lifting nor flame blowout. The flames for the lower left and right hand areas of Fig. 5 surely went through flame mode C, too, but with an excitation level much lower than the indicated and scaled values. For instance, if $S$ and Re$_c$ are, respectively, equal to 0.541 (Re$_a$=1000) and 2800, the resulting flame mode will be flame mode C if the excitation level is tuned more than 20 V. However, if the swirl strength is reduced to 0.49 (Re$_a$=728) while the central jet Reynolds number remains 2800, the flame immediately blows out when the excitation level is 19 V. Such swirl strength ($S$=0.491, Re$_a$=728) is another threshold value for flame stabilization. The flame at this swirl strength and excitation

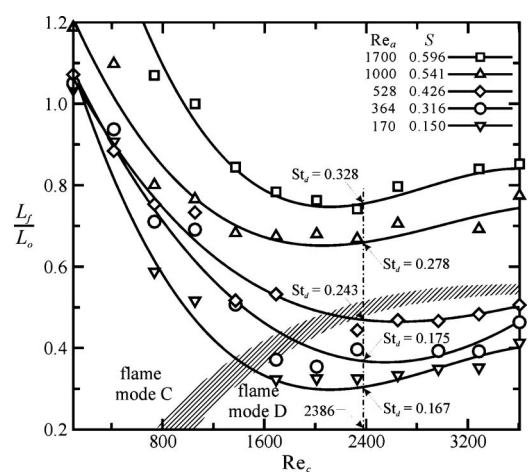


**Fig. 6 Flame length at peak excitation level excited at first fuel tube resonant frequency (*f*=180 Hz)**



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level gets unstable as it tends to lift from the blockage disk. As shown in Fig. 3(*b*), a Strouhal number of 0.257 describes a transition between flame modes C to D.

In contrast to the behaviors of flame shown in Fig. 3(*a*), flames at $\mathrm{Re}_c < 1000$ will exhibit flame elongation at the peak excitation level. This is summarized in Fig. 6; i.e., at $\mathrm{Re}_c = 246$, all variations of swirl strengths ($S = 0.150, 0.316, 0.426, 0.541, 0.596$) showed flame expansion before the flame blows out. However, at a central jet Reynolds number of more than 1000, all conditions of annular swirl strength revealed a flame reduction shown in Fig. 6. The injected volume of fuel in every acoustic cycle for flames at $\mathrm{Re}_c < 1000$ is proportionally larger than its total flame size compared with flames at $\mathrm{Re}_c > 1000$. Thus, the injection time for $\mathrm{Re}_c < 1000$ relative to its volume is longer than the $\mathrm{Re}_c > 1000$ flames. The long injection time will yield a long and *cigarlike* flame, while the shorter injection time guarantees a short *pufflike* buoyant flame, as defined and described by Hermanson et al. [18]. At $\mathrm{Re}_c > 1000$, a least amount of swirl strength has the most reduction in flame length before it blows out. This is due to their differences in the amount of excitation level before flame blows out or reaches the maximum excitation of 25 V. For example, let us compare the flame reduction of the two swirl strengths, namely, 0.316 ($\mathrm{Re}_a = 364$) and 0.541 ($\mathrm{Re}_a = 1000$), at the same $\mathrm{Re}_c$ of 2386. Figure 6 shows an $L_f/L_o$ value for $S = 0.316$ and 0.541 equal to 0.37 and 0.66, respectively. This means that a swirl strength of 0.316 ($\mathrm{Re}_a = 364$) reduces 63% of its original and unexcited flame length before it blows out, while 0.541 ($\mathrm{Re}_a = 1000$) only reduces 34% at the peak excitation level. In Fig. 3, at $\mathrm{Re}_c = 2386$ and $S = 0.316$ ($\mathrm{Re}_a = 364$), the flame blows out when the Strouhal number is less than 0.166 ($E_{\mathrm{exc}} \approx 15$ V). In Fig. 5, at $S = 0.541$ ($\mathrm{Re}_a = 1000$), the flame survives the maximum excitation corresponding to the Strouhal number of approximately 0.308. Thus, for a particular amount of central fuel jet, the least amount of annular flow requires the least amount of excitation level to blow out the flame. It simply shows that the swirl strength has paramount importance in flame stretching and stabilization, which is also implied in Fig. 4(*b*). For $S$ equal to 0.150, 0.316, 0.426, 0.541, and 0.596, the calculated maximum reduction in length respectively occurs at central jet Reynolds numbers of 2182, 2598, 2696, 2124, and 2178. These data were calculated from the fitted equation of the curve for each annular flow Reynolds number in Fig. 6. For a better comparison, 2386 is considered for various annular swirl numbers, particularly in the flow visualization, which will be discussed in the next section. $\mathrm{Re}_c = 2386$ represents an optimum operating condition for various combinations of fuel and swirl jet as far as the present geometry of acoustic excitation is concerned.

**3.4 Flow Patterns and Mixing Mechanics.** The annular and central jets are simultaneously seeded with a 1 $\mu$m diameter MgO seeding particle to trace the flow structures. The Stokes numbers calculated according to Flagan and Seinfeld [25] are 0.02056 and 0.00004, respectively; thus, the seeding particle substantially follows the flow. They are illuminated by a 20 W neodymium doped lithium yttrium fluorides (Nd:YLF) New Wave Pegasus high speed laser. The output green ($\lambda = 527$ nm) laser light beam from such source is polarized to give a 0.5 mm light sheet so that Mie scattering [26] can show sharp plane images. Figure 7(*a*) is randomly chosen from the batch of moving pictures captured by a Sony charge coupled device (CCD) progressive scan camera (DCR-TRV 900NTSC). The frame rate is fixed at 30 frames/s, and the shutter speed is set at 1/1000 s. The purpose of this image is to illustrate the colored interaction of flame between the central and annular flows. As shown in Fig. 7(*a*), a "beadlike" outward rolling vortex is formed at the central jet while the annular jet is influenced by a coherent structure induced by entrainment of the surrounding air. The acoustic excitation is tuned such that the jet exit Strouhal numbers fall between 0.79 and 0.62 in order to understand the mixing mechanics of flow at flame mode A.

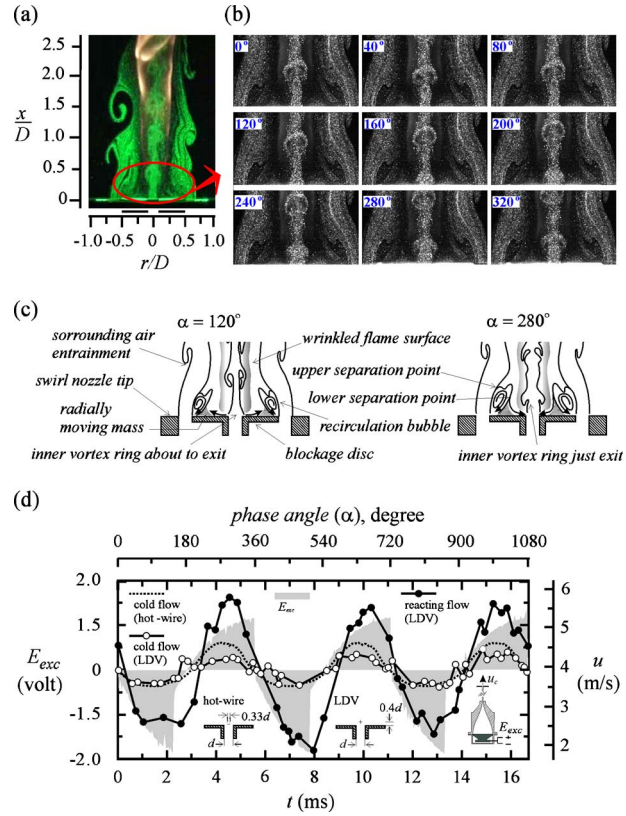


**Fig. 7 Mixing mechanism of flame mode A at $\mathrm{Re}_c = 2386$, $\mathrm{Re}_a = 364$ ($S = 0.316$), $f = 180$ Hz, and $0.79 > \mathrm{St}_d > 0.62$. (*a*) Near field image of flame and flow; shutter speed is 1/1000 s. (*b*) Chronological images of flame and flow. (*c*) Elucidations of flow feature. (*d*) Central jet velocity fluctuations and phase angle.**

The evolution of these beadlike structures of central flow in flame mode A has been given special treatment by using an IDT XS-4 high speed camera. It has a maximum framing rate of 5000 frames/s at a resolution of $512 \times 512$ pixels. The frame rate of images shown in Fig. 7(*b*) is 3240 pictures/s so that 18 equal images will be captured in every acoustic cycle of 180 Hz. A common special feature in every cycle, which transpired from the total motion pictures, is elucidated in Fig. 7(*c*). Such sequence of images are being phase locked with the flow by a small background light emitting diode (LED) light, which glowed during the positive half of the acoustic cycle. The LED's emitted light is positioned at the lower left most corner of every image being captured by the camera together with the flow. The function generator signal that powers the background LED light is also the reference signal in velocity measurement. The same signal is amplified to power the loud speaker embedded at the bottom of the fuel tube. They are captured simultaneously by a Keithley's KUSB-3116 high speed data acquisition system; thus, a phase angle of each and every image acquired is established and shown in Fig. 7(*d*). The start of every acoustic cycle is arbitrarily chosen at 0 V when the face of the loud speaker goes down. Although the excitation voltage from the function generator is a square wave signal, it is distorted after being amplified to power the loud speaker. Every acoustic cycle is composed of a 2 ms negative half followed by a 3.55 ms positive half, which completes the cycle.

As shown in Fig. 7(*d*), the jet exit accelerates during the second and third quarter of the acoustic cycle. The jet acceleration can also be validated by the elucidation of Fig. 7(*c*) where the fuel vortex clearly comes out from the jet exit at phase angles of 120–300°. The fuel vortex ring rolled outward to entrain oxidizing air, thereby increasing its volume as it moved downstream [17]. The



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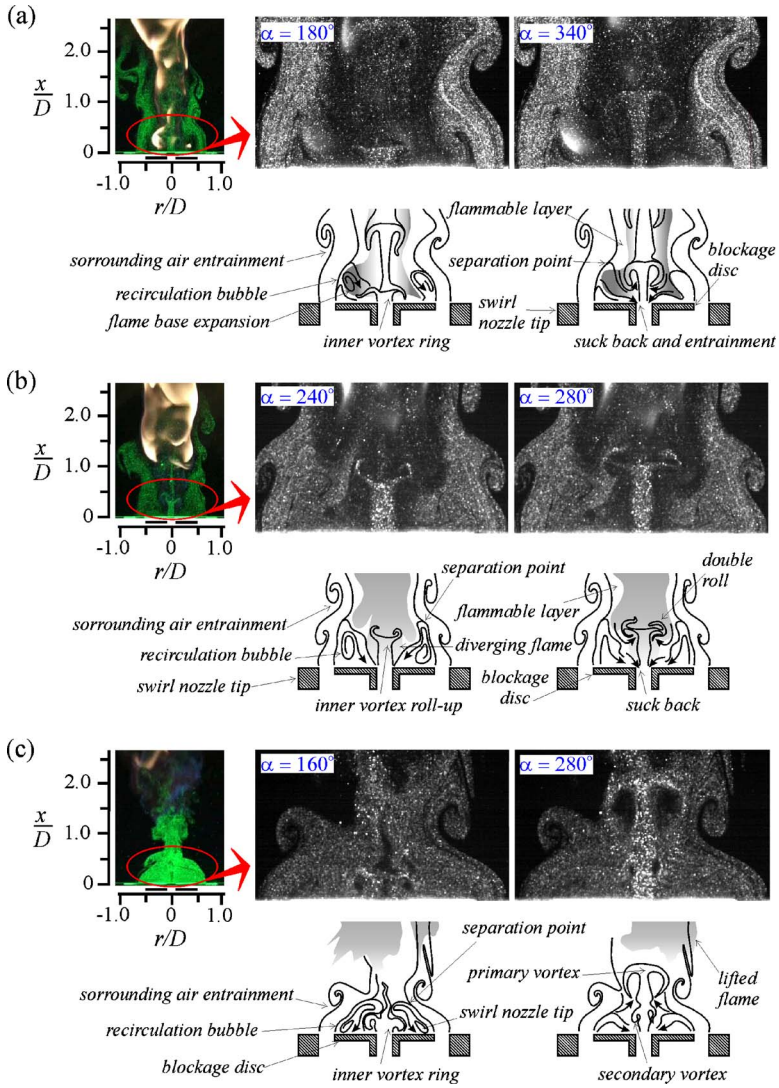


**Fig. 8 Mixing mechanism of flame modes B, C, and D with jet exit Strouhal numbers respectively equal to (*a*) 0.44 ($E_{exc}$=2 V), (*b*) 0.29 ($E_{exc}$=4 V), and (*c*) 0.19 ($E_{exc}$=8 V). $Re_c$=2386, $S$=0.316 ($Re_a$=364), and $f$=180 Hz. Colored picture shutter speed, 1/1000 s.**

processing of this jet vortex necessarily wrinkled the flame surface at flame mode A, as discussed in Sec. 3.2. A larger amount of annular air is entrained and causes some flame structures to pinch off at near downstream sections [6]. Previous work [3,4,23] illustrated a dual ring setting in the blockage disk: an outer ring, which is presently labeled as a recirculation bubble, and an inner ring, which decomposes into a radially moving mass due to acoustic pulsing. As shown in Fig. 7(*b*) and elucidated in Fig. 7(*c*), a radially moving mass cannot be transformed into an inner vortex ring because it is consumed by the central fuel jet during the positive half of the acoustic cycle. The radially moving mass absorbed the heat radiated to the blockage disk and consequently preheated the oxidizing air. This moving mass is displaced by the fuel vortex during the second and third quarter of the acoustic cycle. At this period, a large quantity of fluid escaped through the upper separation point of the recirculation bubble. These periodic exchanges of momentum, accelerating and decelerating the central fuel jet, substantially enhanced turbulence and mixing [13]. As the fuel vortex left from the jet exit during the fourth to the first quarter of the next acoustic cycle, the radially moving mass is entangled by the central jet through the lower separation point. At this instant, the lower separation point of the recirculation bubble displaced a large quantity of preheated oxidizing gas to the central jet, which is very useful for spontaneous combustion.

Figure 8 shows how the flow structure is dramatically altered by a large acoustic excitation level. The chosen images of flow in Figs. 8(*a*)–8(*c*) are taken during the third quarter of the acoustic cycle where the fuel vortex has totally ejected but is still visually identifiable. The region of interest in each of the colored pictures is zoomed out by the corresponding images from the black and white pictures taken by a high speed camera. The flow features are also respectively elucidated directly under each of these motion pictures. As shown in Fig. 8(*a*), at a Strouhal number of 0.44, the fuel vortex obviously grows larger compared with Fig. 7. The fuel jet separated from the wall of the fuel tube, as elucidated at $\alpha$ =340 deg; thus, the potential core shrinks to allow the surrounding fluid to be sucked into the fuel tube [7,16]. This makes an in-tube burning [9] near the wall where the central portion remains a rich mixture, which rolls up wide to the recirculation bubble, as substantially shown in the elucidation of $\alpha$=180 deg (Fig. 8(*a*)). Consequently, a flammable lean mixture is achieved



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on top of the blockage disk. The periodic flaming of the recirculation bubble made the total reaction zone expanded at its base described in Sec. 3.2 as flame mode B. The radially moving mass, however, disappeared because it is totally consumed by the central jet as the pulsing amplitude necessarily escalates the entrainment of such surrounding gas [7,16,17]. Entrainment fathomed down to the fuel tube as excitation increased and tuned the flame to mode C. Figure 8(*b*) is an example of flame and flow interactions that caused the flame base to be diverged. The central jet Strouhal number is 0.29, which represents flame mode C. The roll-up of vortex shown in Fig. 8(*b*) is caused by the entrainment momentum, which is pushed by the incoming acceleration of the fuel jet. In the case of pure jet excitation where flame is not stabilized by swirl flow, the rolling up of this vortex will block the entrainment of surrounding air, causing the flame to elongate [8]. For the case of the present work, certainly it moved opposite with the rim of the recirculation bubble catching a sufficient amount of oxidizing air and preventing the flame from elongation. However, the effect remains obvious as the flame length reduction normalized compared with flame modes A and B described in Sec 3.2. The roll-up vortex is initially composed of a lean mixture left behind from the previous acoustic cycle. The strong magnitude of the recirculation bubble shattered the upward rotating vortex, generating another roll-down vortex behind it, as shown in the elucidation of $\alpha = 280$ deg in Fig. 8(*b*). This back-to-back and double rolled central vortex arose at approximately the same level with the upper separation point of the recirculation bubble in the downstream area. Some portions of the burned gases, together with the preheated circulating air at the recirculation bubble, are sucked back to the fuel tube and ignited the collapsible mixing zone. Increasing the excitation level entice more gas to be entrained deep down the fuel tube, resulting in slight flame length reduction. The transition of this mixing mechanism is efficiently transferred to the next acoustic cycle validated by blue divergent flame continuously anchored at the tip of the fuel tube. But if the excitation level reached its peak value, such mixing transition became discontinuous. A periodic absent of flammable mixture at the tip of the fuel tube lifted the flame to the downstream area. Severe acceleration of fuel and the insufficient quality of premixing caused flame blowout especially at low swirl strength. Increasing the swirl strength consequently improved the shear stress between the fuel vortex and the recirculation bubble, preventing the flame from blowing out, as implied by Figs. 5 and 6. The flow phenomena of a lifted flame are described with the aid of Fig. 8(*c*), which demonstrates the periodic capping of the fuel tube at the blockage disk. It is due to the periodic absence of the fuel jet during a high excitation amplitude. The Strouhal number at the jet exit is 0.19, and the flame is lifted, which presents flame mode D, as shown in Fig. 3(*b*). As shown in Fig. 8(*c*), a large primary or leading fuel vortex penetrates the recirculation bubble and leads ahead of the group of secondary vortices, which Gharib et al. [27] called *the trailing jet of fluid*. They are separated by the so called pinch-off process [27] by which the leading vortex ceased to entrain an additional mass of fluid and continue to circulate to the downstream section. Considering the distance traveled by the primary and secondary vortices and the amount of oxidizing gas being entrained to the mixture, premixing is appreciably enhanced [11,12,14] better than the other flame modes. This can be validated by the flame length shown in Fig. 3 where a lifted flame or flame mode D showed satisfactory combustion performance in terms of flame color and length. The primary or leading vortex therefore initially bears the right stoichiometric ratio by premixing with the entrained fluid in the fuel tube. Excessive entrainment by a large excitation level therefore pushes the mixture of the leading vortex to a lean flammable level, which will result to the failure of spontaneous ignition. This is, of course, combined with the excessive strain brought by the accelerating fuel vortex. Whichever is the dominant of the two aforementioned processes that cause the flame blowout is beyond the scope of this work.

## 4 Conclusions

A nonpremixed reacting swirling wake has been excited acoustically at various ranges of excitation levels and frequencies. The fuel tube resonance frequency was experimentally obtained at 180 Hz and its harmonics by which minimal excitation level can modify the flame. Velocity at the jet exit was measured by two-component LDV at the first fuel tube resonant frequency. The Strouhal number of the acoustic excitation was calculated based on the first fuel tube resonant frequency and the root mean square of the fluctuating velocity at the central jet exit. Swirl strength substantially damped the jet velocity fluctuation and at the same time mitigated the reduction in flame length. In the summarized representation of flame reduction, the forcing system effectively reduced the flame length at $\mathrm{Re}_c = 2386$. Flammability and stability chart illustrated that flame stabilization is weak at a swirl strength close to 0.491 ($\mathrm{Re}_a = 728$).

The flame at the swirling wake exhibited four different flame behaviors during acoustic excitation classified into flame modes A, B, C, and D. The mechanics of mixing at each flame mode was discerned from phase locked Mie scattering images captured by a high speed camera. At mode A, low excitation amplitudes induce beadlike vortices in the central fuel jet and make the flame wrinkles. At mode B, the central jet vortices become larger with the increase in excitation amplitude and thus lead to a wider and shorter flame. At mode C, the excitation amplitude is increased further to exceed the critical value so that the central jet vortices change the rotation direction and pacing with the annular jet vortices. Large turbulent intensity and mixing are induced due to the drastic change in flow field and therefore make the flame looks blue and short. At mode D, a lifted flame is formed because the excitation amplitude is large enough to modify a strong but small recirculation bubble sitting on the disk.

## Nomenclature

$B$ = blockage ratio ($=D^2/D_o^2$), dimensionless
$D$ = diameter of circular blockage disk, mm
$d$ = diameter of central hole at circular blockage disk where the central jet goes out, mm
$D_h$ = hydraulic diameter of annular swirling jet at exit ($=D_o-D$), mm
$D_m$ = mean diameter for the calculation of swirl number ($=(D+D_o)/2$), mm
$D_o$ = outer diameter of annular swirling jet at exit, mm
$f$ = excitation or forcing frequency from the function generator, Hz
$L_f$ = length of a flame during acoustic excitation, mm
$L_o$ = length of a flame prior to acoustic excitation, mm
$E_{\mathrm{exc}}$ = root mean square excitation voltage measured at the loud speaker terminals, V
$\mathrm{Re}_a$ = exit Reynolds number of annular flow, Eq. (2), dimensionless
$\mathrm{Re}_c$ = exit Reynolds number of central jet, Eq. (3), dimensionless
$r$ = radial coordinate, originated from center of circular disk, mm
$S$ = swirl number, Eq. (1), dimensionless
$\mathrm{St}_d$ = central jet exit Strouhal number, Eq. (4), dimensionless
$t$ = time, ms
$u$ = axial velocity component, m/s
$u_a$ = volumetric mean axial velocity of annular swirling jet at exit, m/s
$u_c$ = volumetric mean axial velocity of central jet at exit, m/s



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$u_{rms}$ = root mean square velocity at the jet exit during acoustic excitation, m/s
$w$ = azimuthal velocity component, m/s
$x$ = axial coordinate, originated from center of circular disk, mm

## Greek Symbol

$\alpha$ = acoustic excitation phase angle, deg
$\nu_a$ = kinematic viscosity of air, $m^2/s$
$\nu_f$ = kinematic viscosity of fuel, $m^2/s$
$\lambda$ = wavelength of the laser light beam, nm
$\delta$ = dimension of LDV's measuring volume, mm

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